Q5. Operating system and network information

Original question: Use basic Linux commands to display the operating system information, current date, command history, and network interface details. Record the operating system name and IP address of the Kali Linux machine.

Aim

Display OS details, current time, command history and network interfaces, and record Kali's actual local IP address.

Requirements

An interactive Kali/Linux terminal. Type `bash` first if you need the Bash-specific history form below. No internet or root access is needed.

Procedure

Step 1 - Display OS and time

```
mkdir -p "$HOME/EHF_Exam/Q5"
cd "$HOME/EHF_Exam/Q5"
cat /etc/os-release | tee os-release.txt
uname -a | tee kernel.txt
date -Is | tee date.txt
```

Read PRETTY_NAME and NAME in os-release for the distribution. `uname -a` reports kernel/system information, not the distribution's marketing release name. `date -Is` includes the timezone offset; the clock may be wrong if not synchronized.

Step 2 - Display history

```
history | tail -n 20 | tee history_last20.txt
```

`history` is a shell builtin; `tail` keeps the final 20 entries. History may be empty in a new Bash session, disabled, or limited to commands typed in that shell. In Zsh, use `fc -l -20` if remaining in Zsh. Never publish history containing credentials; do not enter secrets as command arguments.

Step 3 - Inspect interfaces, addresses and routes

```
ip -br link | tee interfaces.txt
ip -br addr | tee addresses.txt
ip route | tee routes.txt
hostname -I
```

`ip -br` gives brief output. `link` shows link state; `addr` shows IPv4/IPv6 addresses; `route` shows IPv4 routes. `hostname -I` is a compact address list and may show more than one interface. For a known lab destination, `ip route get <LAB_TARGET_IP>` (replace the placeholder) shows the selected source address without scanning it.

Expected output - illustrative

```
PRETTY_NAME="Kali GNU/Linux Rolling"
lo      UNKNOWN  127.0.0.1/8 ::1/128
eth0    UP       192.0.2.10/24
```

192.0.2.10 is a documentation address, not your IP. Interface names can be eth0, wlan0, enp0s3, tun0 or others. `/24` is a prefix length, not part of the address to enter as a scan target.

Interpretation, observation and result

Record distribution name, kernel, date/time/timezone, active lab interface, IPv4/prefix and relevant IPv6 addresses. Do not select 127.0.0.1 as the machine's lab address: it is loopback. With NAT, a VPN or multiple adapters, choose the interface/source actually used for the lab network. These commands generally do not reveal the internet-facing public IP behind NAT. The required system and local network details are documented once the saved outputs agree with your chosen interface.

Common errors

- No IPv4 on the lab adapter: check link state, DHCP/static configuration and VM attachment.
- ifconfig missing: use ip; installing legacy net-tools is unnecessary.
- History missing: demonstrate a few harmless commands in the current interactive shell and retry.

Viva questions

- Does uname identify the distribution reliably? It identifies the kernel; use os-release for the distribution.
- What is 127.0.0.1? The IPv4 loopback address.
- Can a machine have several IP addresses? Yes, across interfaces or on one interface.
- Why record timezone? It makes timestamps interpretable and comparable.